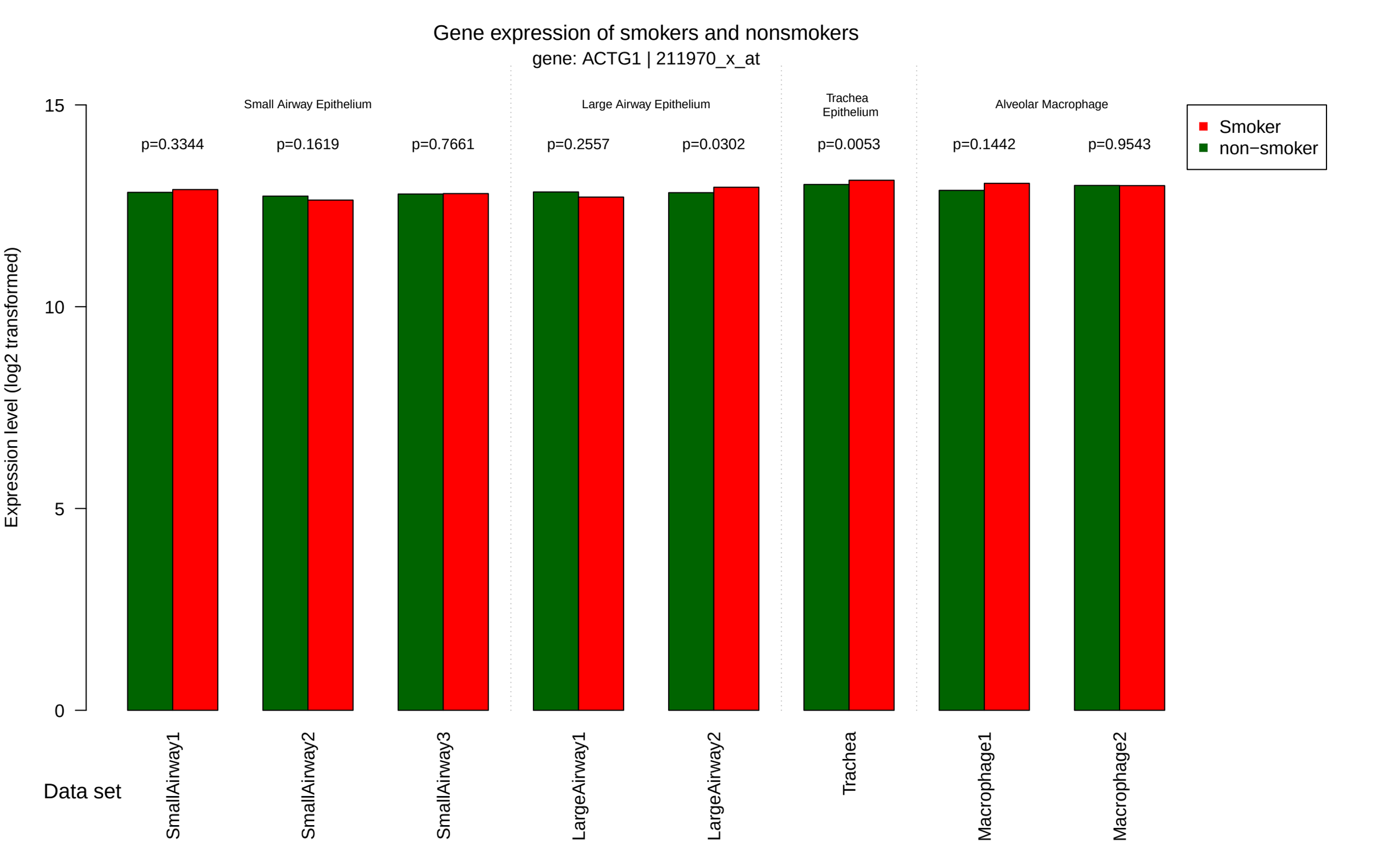
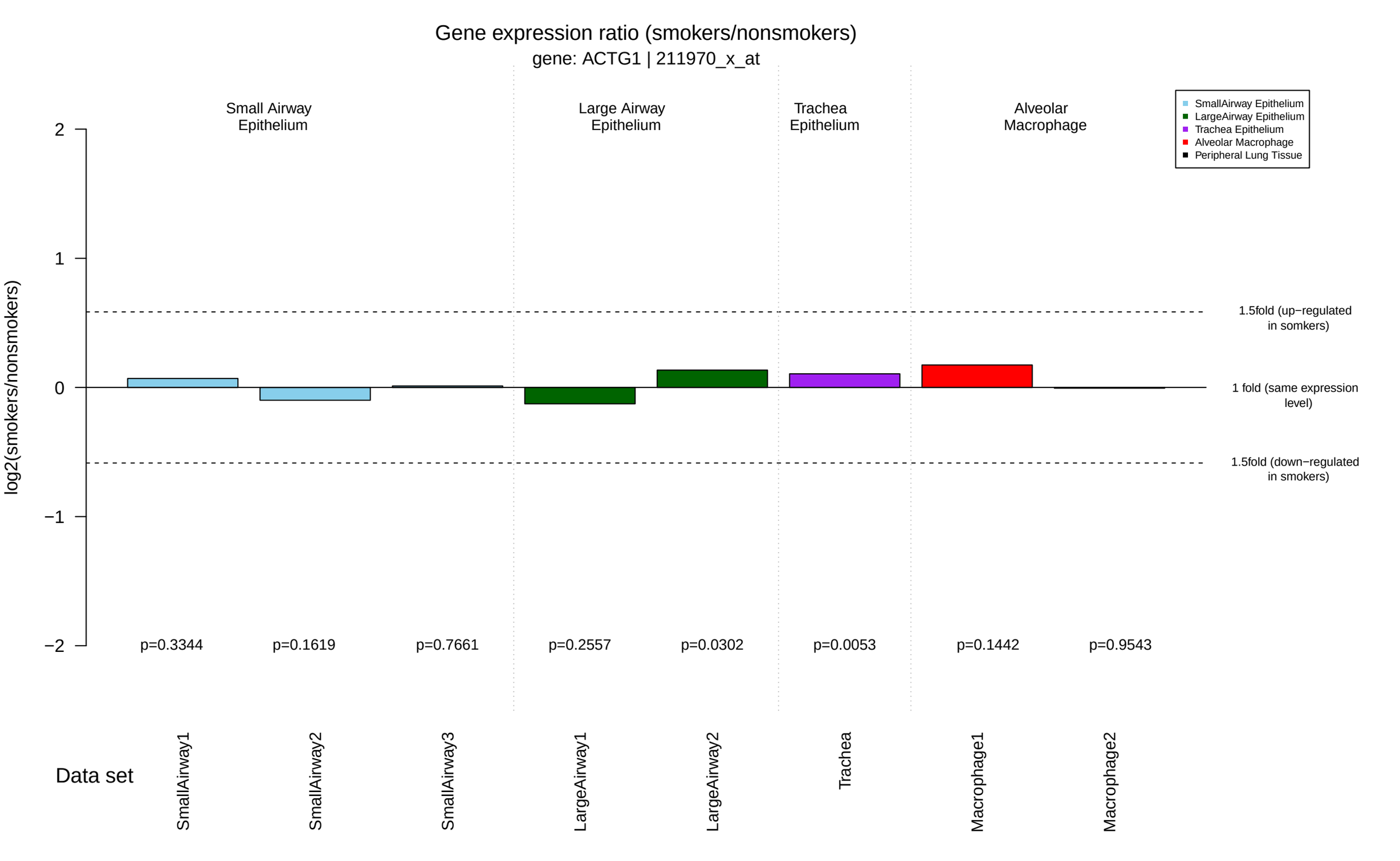
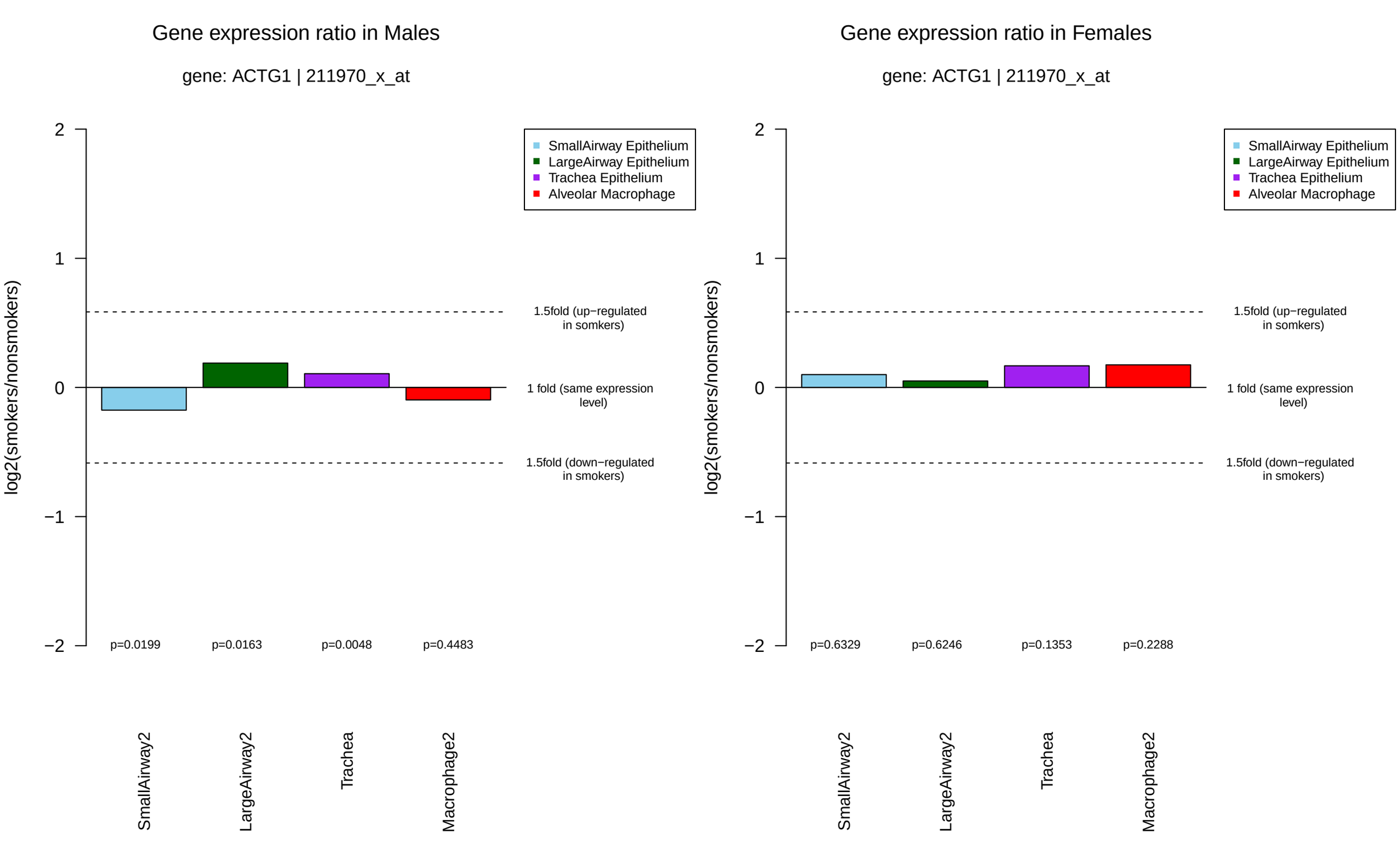
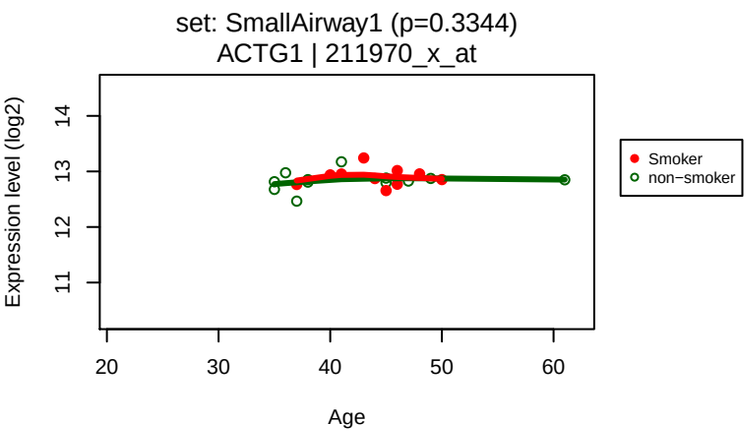
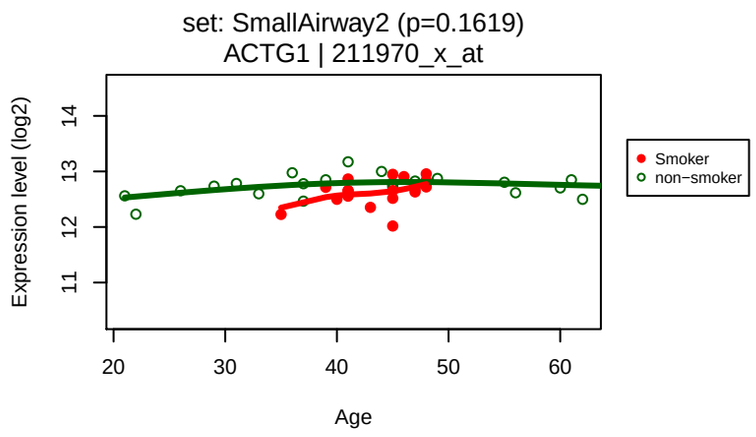
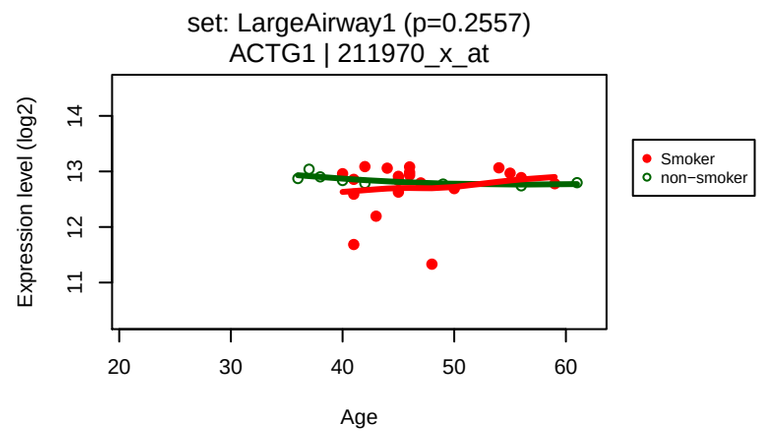

Gene expression of smokers and nonsmokers
gene: ACTG1 | 211970_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3344
p=0.1619
p=0.7661
p=0.2557
p=0.0302
p=0.0053
p=0.1442
p=0.9543
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

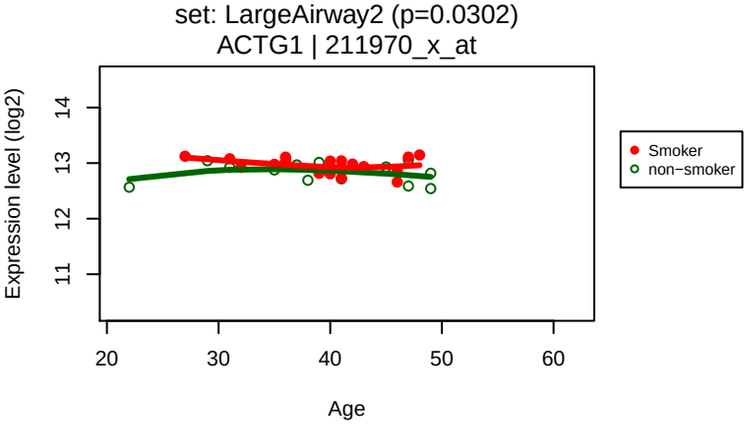

Gene expression ratio (smokers/nonsmokers)
gene: ACTG1 | 211970_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3344
p=0.1619
p=0.7661
p=0.2557
p=0.0302
p=0.0053
p=0.1442
p=0.9543
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

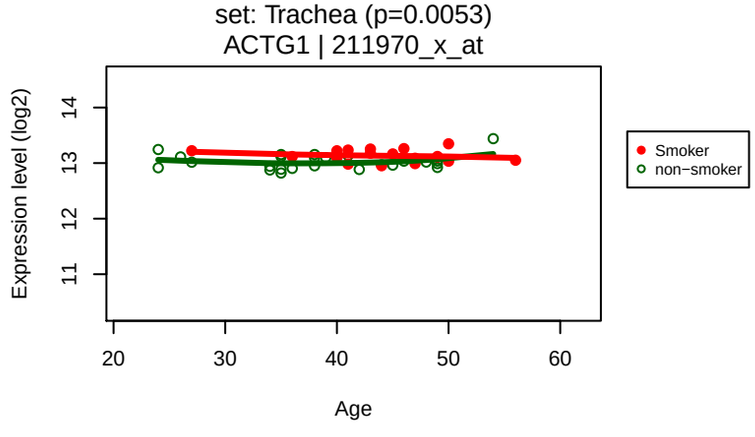

Gene expression ratio in Males
gene: ACTG1 | 211970_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0199
p=0.0163
p=0.0048
p=0.4483
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTG1 | 211970_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6329
p=0.6246
p=0.1353
p=0.2288
SmallAirway2
LargeAirway2
Trachea
Macrophage2

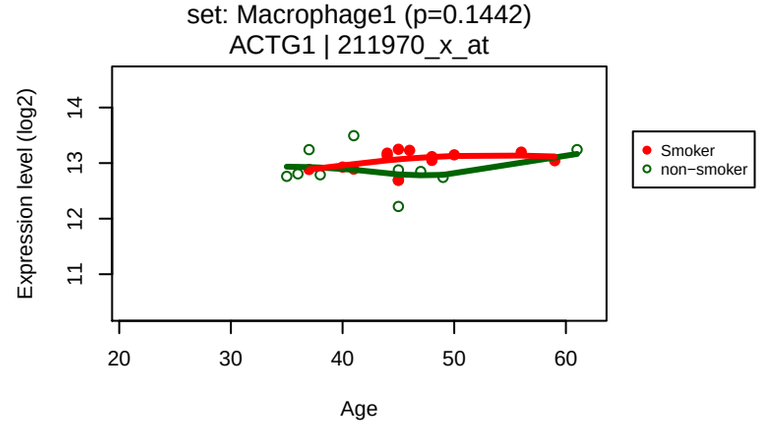

set: SmallAirway1 (p=0.3344)
ACTG1 | 211970_x_at
Expression level (log2)
Age
Smoker
non−smoker

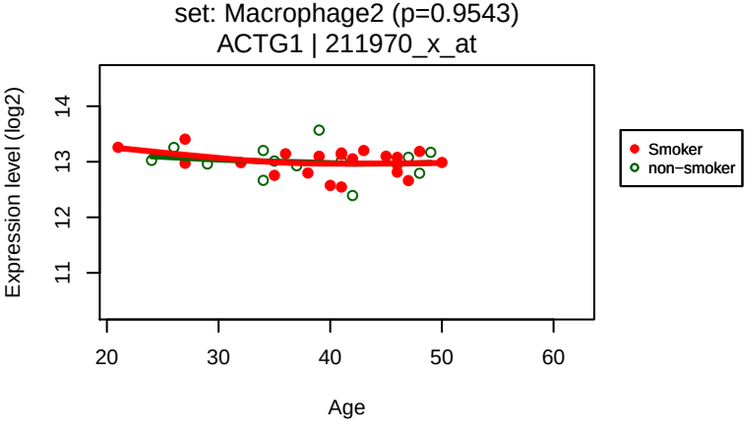

set: SmallAirway2 (p=0.1619)
ACTG1 | 211970_x_at
Expression level (log2)
Age
Smoker
non−smoker

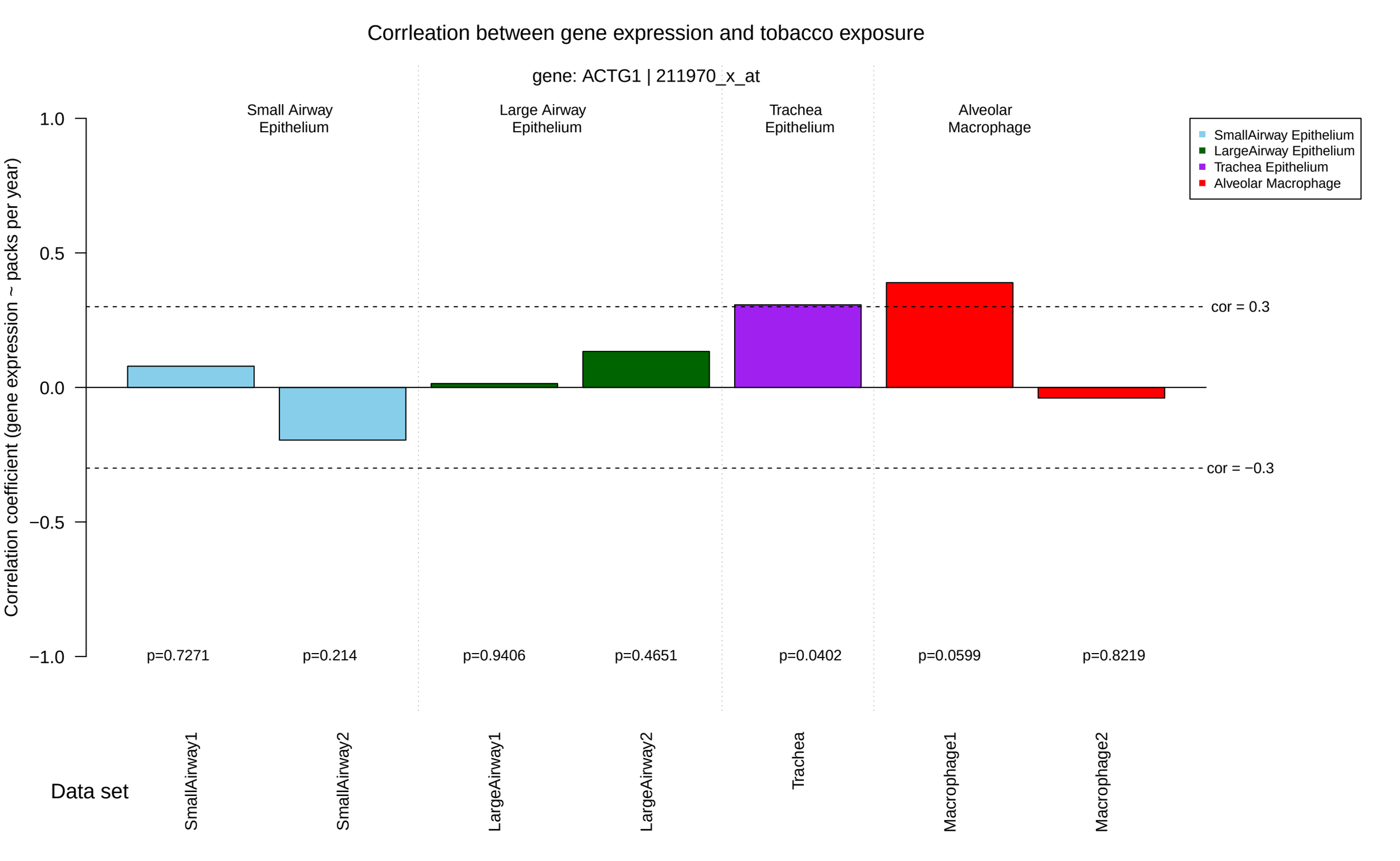

set: LargeAirway1 (p=0.2557)
ACTG1 | 211970_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0302)
ACTG1 | 211970_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0053)
ACTG1 | 211970_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1442)
ACTG1 | 211970_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9543)
ACTG1 | 211970_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTG1 | 211970_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7271
p=0.214
p=0.9406
p=0.4651
p=0.0402
p=0.0599
p=0.8219
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2